

Data Card

Team Gambia — TRI AI Saturdays Cohort 10 — Agricultural Extension RAG: Smart Retrieval for Farmers

Dataset Summary

Source	Kaggle competition: "Agricultural Extension RAG: Smart Retrieval for Farmers" (host: TRI AI)
Corpus size	695 agricultural extension factsheets
Train queries	308 farmer questions, with 4,194 graded relevance judgements
Test queries	200 farmer questions (public + private leaderboard, unlabeled to participants)
Task	Document retrieval / ranking (top-5 per query)
Metric	nDCG@5 (Normalized Discounted Cumulative Gain at rank 5)

Composition

The knowledge base spans 12 named crops (maize, tomato, rice, cassava, common bean, cowpea, groundnut, sorghum, plantain, yam, cocoa, pearl millet) plus general-topic documents (soil management, climate adaptation, drought). Source attribution spans multiple agricultural research and extension bodies, including FAO, CGIAR, IITA, ICRISAT, Plantwise, AGRA, and national extension services.

Documents are drawn from two origins: the majority (637 of 695) are synthetic, CC0-licensed factsheets; the remainder are LLM-grounded rewrites of open-access, CC-BY-licensed source material. This mix was a deliberate design choice by the dataset creators to introduce realistic variation in vocabulary and phrasing style across the corpus.

Relevance Labeling Scheme

Each training query is associated with a set of graded relevance judgements on a 0–3 scale:

3 — Perfect	Directly and completely answers the question.
2 — Relevant	A closely related aspect of the same topic.
1 — Marginal	The same underlying issue, but on a different crop.
0 — Hard negative	Not relevant, including documents deliberately constructed to be lexically or topically similar but fa

On average, each training query has approximately 13.6 judged documents and 1.8 documents with the highest relevance grade, meaning most questions have more than one acceptable answer — retrieval quality depends on ranking order, not just finding any one correct document.

Known Limitations and Biases

- The corpus is weighted toward certain crops (Maize: 128 documents, Tomato: 102, Rice: 78) versus others (Pearl millet: 10), which may bias retrieval performance unevenly across crop types.
- Synthetic construction of the majority of documents, while controlled and license-clean, may not fully capture the linguistic variation of authentic field-collected extension material.
- Query phrasing in the training set follows a small number of recurring templates (e.g. "How do I manage X in Y?" / "What should I do about an outbreak of X in Y?"), which does not necessarily reflect the full diversity of real farmer phrasing in deployment.
- The dataset intentionally includes hard negatives constructed to penalize naive lexical retrieval, which means baseline keyword-matching approaches will systematically underperform by design — this is a deliberate evaluation choice, not an accidental artifact.

Intended Use

This dataset is intended for benchmarking and developing retrieval components of agricultural RAG systems. It is not intended, on its own, to be deployed directly to farmers without a downstream generation and safety-review layer, and should not be treated as a substitute for professional agronomic or veterinary advice.